

Course Code	EC24P
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Class	T.E – EXTC. SEM-V
Division	B
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EXPERIMENT NO.4

Estimated time to complete this experiment: 02 hours.

Objective: To plot the VTC of CMOS Inverter using gpgk 180.

CO to be achieved: CO2.

Expected Outcome of Experiment:

- Students will learn cadence virtuoso.
- Students will be able to plot VTC of CMOS inverter using CADENCE Virtuoso.
- Students will be able to calculate Noise margin and Peak power consumption of CMOS inverter.

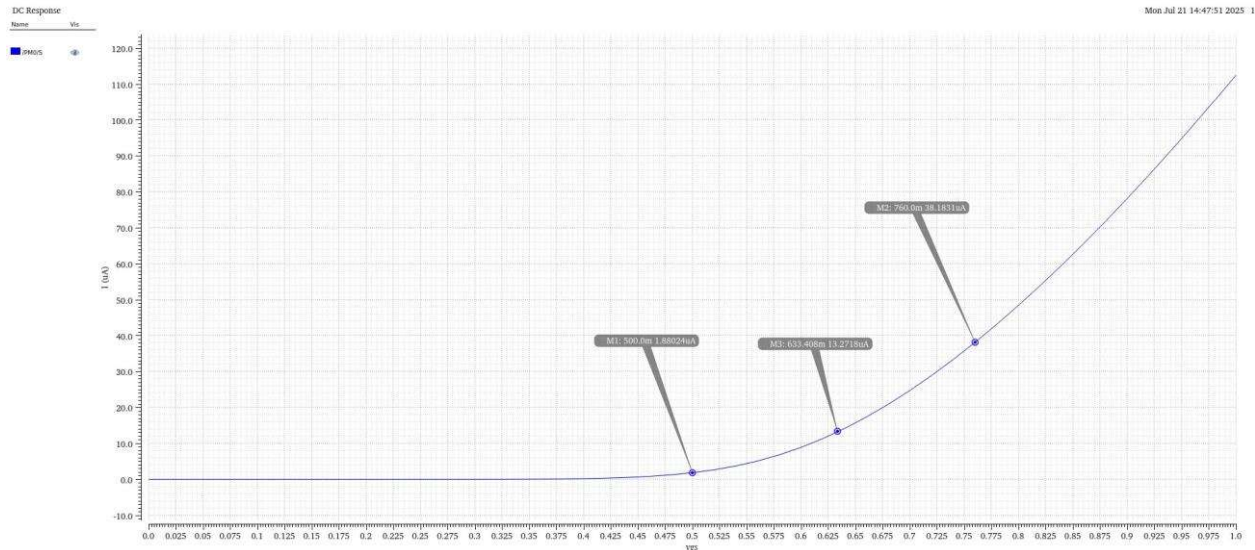
Pre Lab/ Prior Concept: Knowledge of working of MOSFET and CMOS Inverter.

Apparatus/TOOL:

- PC

- CADENCE VIRTUOSO.

Circuit Diagram:



5) Procedure.

- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
 - csh
 - source /home/install/cshrc
 - virtuoso.
- Create the Library and cell view
- Open log window using terminal.
- Click on the new option, now create library with gpd180 from technology library.
- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.

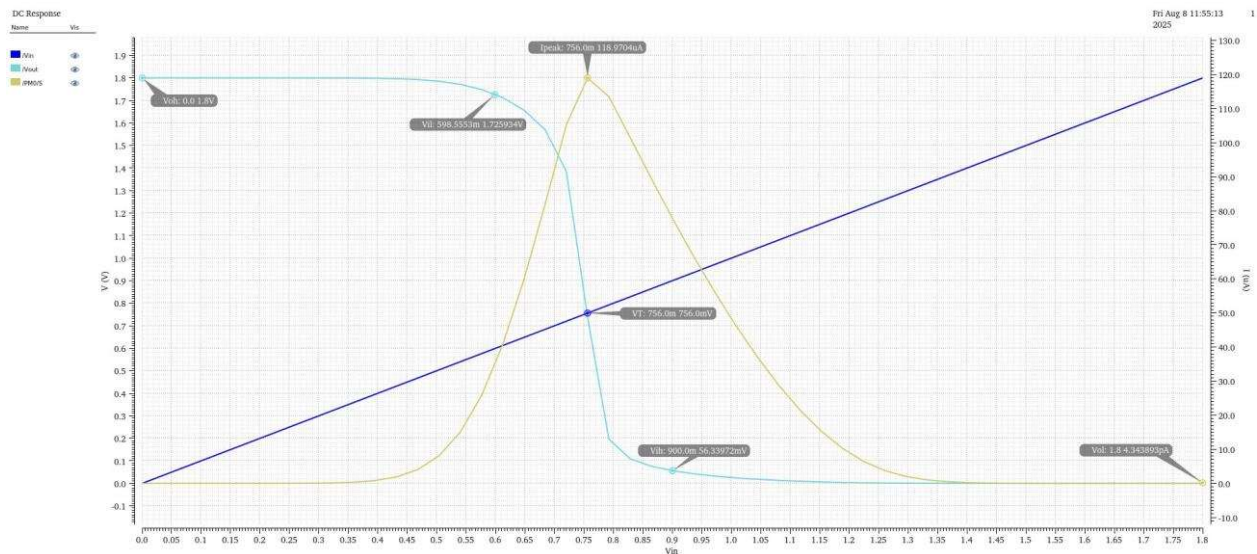
For characterization of 180nm CMOS Inverter.

- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as W 1um (micro-meter).
- Select PMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as W 1um (micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.
- Take VDD=1.8V and VGG=v_{gg} V.
- Then plot the Voltage Transfer Characteristics of CMOS Inverter.
- Then from the VTC graph find the following values.
 - V_{IL}
 - V_{OH}
 - V_{OL}

- V_{IH}
- V_{TH}
- I_{PEAK}

➤ Using above values calculate the Noise Margin and Peak Power of CMOS Inverter.

Results:



Calculation:

$$\begin{aligned}
 N_{MH} &= V_{OH} - V_{IH} \\
 N_{ML} &= V_{IL} - V_{OL} \\
 N_{MH} &= 1.8 - 56 \text{ mV} \\
 &= \underline{1.744 \text{ V}} \\
 N_{ML} &= 1.725 \text{ V} - 1.435 \text{ mV} \\
 &= \underline{1.724 \text{ V}} \\
 \text{Peak Power} &= V_{DD} \times I_{PEAK} \\
 &= \underline{214.146 \text{ } \mu\text{V}}
 \end{aligned}$$

Conclusion:

In this experiment, we successfully plotted the Voltage Transfer Characteristics (VTC) of a CMOS inverter using **Cadence Virtuoso**. We also calculated important parameters such as **VIL, VIH, VOH, VOL, VTH**,

and used them to find **Noise Margins** and **Peak Power Consumption**. This helped us understand how a CMOS inverter works, its switching behavior, and its reliability in digital circuits.

Post lab Questions.

Q1) What is the significance of Noise Margin?

Noise margin shows how much noise a digital circuit can tolerate without affecting its correct operation. Higher noise margin means better reliability.

Q2) In CMOS Inverter why PMOS is used as pull up and NMOS is used as pull down?

PMOS passes a strong logic '1' and weak logic '0', so it is used to pull the output high. NMOS passes a strong logic '0' and weak logic '1', so it is used to pull the output low. This combination gives low power consumption and proper logic levels.

Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.